

Allu Yasoda Kishore

Machine Learning Engineer · AI Systems · Full-Stack Development

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PROFESSIONAL SUMMARY

I build and ship machine learning systems end to end, from designing text and image classification models through training, validation, and deploying them to production using Python, TensorFlow, and scikit-learn. I care about the full lifecycle, including experimentation and reproducibility as well as monitoring and continuous improvement once something is live. I have built data pipelines for preprocessing large-scale datasets, deployed services with FastAPI and AWS Lambda, and worked on a published research project applying interactive 3D avatars and large language models to behavioral support in digital games.

EDUCATION

Bachelor's Qualification Plan in Computer Science 2025 – 2026
Blekinge Institute of Technology (BTH) Karlskrona, Sweden

Bachelor of Technology – Computer Science and Engineering 2022-2025
University College of Engineering, Kakinada Kakinada, India
CGPA: 8.3 · Coursework: Artificial Intelligence, Data Analytics, Machine Learning, NLP, Computer Networks

Intermediate – MPC (Mathematics, Physics, Chemistry) 2020 – 2022
Aditya Junior College Andhra Pradesh, India
Score: 91.3%

SSC – Secondary School Certificate 2020
Bhashyam English Medium School Andhra Pradesh, India
Score: 93%

PROFESSIONAL EXPERIENCE

Backend AI Engineering Intern – FlyRank AI June 2026 - August 2026

- Built and deployed ML models end to end covering design, training, validation, and production deployment using Python, TensorFlow, and PyTorch, taking full ownership of the lifecycle from experimentation through monitoring
- Developed FastAPI services to expose ML models as production APIs, ensuring reliable and observable systems that scaled under load with proper monitoring and error handling in place
- Built data pipelines for preprocessing and transforming datasets, applying best practices in experimentation, reproducibility, and testing throughout the development process

Machine Learning Intern – Cognifyz Technologies Jul 2025 – Aug 2025

- Designed and trained a cuisine classification model achieving 96% accuracy on a 70,000-plus restaurant dataset using scikit-learn and Pandas, taking ownership of the project from experimentation through deployment
- Built a restaurant rating prediction model achieving an R squared of 0.9859, with RMSE of 0.1792, MAE of 0.1210, and MAPE of 3.74%, on a large-scale production dataset
- Developed a content-based restaurant recommendation system and an interactive Streamlit analytics dashboard for location-based trends
- Built scalable data pipelines using Pandas and SQL for preprocessing, validation, and transformation of high-volume restaurant data, deploying models on AWS with monitoring for continuous improvement

AI Intern – Intrainz Mar 2024 – May 2024

- Developed image classification and NLP pipelines using TensorFlow and PyTorch, taking ownership of the full ML lifecycle from problem definition through production deployment and monitoring
- Collaborated with engineering teams to build and maintain production ML systems, ensuring reproducibility, reliability, and observability in high-volume environments
- Implemented automated testing and monitoring for model performance, setting up systems to catch degradation and enable quick rollback when needed

AI Intern – EiSystems May 2024 – Jun 2024

- Built and deployed ML classification systems in production using Python and scikit-learn, managing the full lifecycle from training through deployment and continuous monitoring
- Designed data processing pipelines and implemented best practices in experimentation and documentation, enabling team reproducibility and knowledge transfer
- Worked autonomously on system design and scalability challenges, collaborating with teams to explain technical decisions and trade-offs clearly

Cloud Computing Intern – DevSkillsHub, India

April 2024 - May 2024

- Designed and deployed cloud-based applications using AWS services such as S3, EC2, and Lambda to support scalable data processing
- Built and managed data pipelines for data ingestion and transformation, and implemented serverless architectures to optimize performance and cost
- Developed REST APIs for integration and monitored system performance using cloud monitoring tools, following best practices for scalability and security

Student Graduate Program – Wipro, India

2023

- Completed a structured industry training program focused on artificial intelligence, full-stack web development, and software engineering practices
- Designed and implemented machine learning models using data preprocessing, feature engineering, and performance evaluation techniques
- Developed and deployed responsive web applications with integrated frontend and backend components, applying analytical problem-solving in team-based environments

RESEARCH PUBLICATIONS

[1] Y. K. Allu et al., “Does Embodiment Matter? Evaluating AI Avatars for CBT-Inspired Interventions in Games,” accepted at IEEE GEM 2026, session GEM#7: Game AI, Generation and Development Tools, Berlin, Germany, September 2026. (DOI pending)

Presents a system integrating a locally hosted Llama-3.2-3B-Instruct language model with Unity-based 3D avatar rendering, real-time speech synthesis, and WebSocket communication to deliver Cognitive Behavioral Therapy inspired interventions through interactive avatars in digital games. Includes an empirical user study evaluating engagement, perceived support, and communication effectiveness.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, Bash, C# (Unity)
- **ML Frameworks:** TensorFlow, PyTorch, scikit-learn, Llama LLMs
- **API Development:** FastAPI, REST API design, WebSocket communication
- **Data Processing:** Spark, Databricks, Pandas, NumPy, SQL
- **ML Specialisations:** Text classification, image classification, NLP, speech-to-text, LLM integration
- **3D and Game Development:** Unity, 3D avatar animation, real-time rendering, interactive systems
- **Cloud and Infrastructure:** AWS (S3, EC2, Lambda), Docker, Kubernetes
- **ML Lifecycle:** Experimentation, testing, monitoring, observability, reproducibility, user study evaluation
- **Version Control:** Git, GitHub

KEY PROJECTS

Impact of Interactive 3D Avatars for CBT-Inspired Behavioral Support to Players in Digital Games

Bachelor's thesis at BTH, accepted as an IEEE GEM 2026 paper, exploring the effectiveness of 3D avatars delivering Cognitive Behavioral Therapy interventions in digital game environments.

- Designed and implemented a full end-to-end system integrating a locally hosted Llama-3.2-3B-Instruct LLM back-end with Unity-based 3D avatar rendering, real-time speech synthesis via Microsoft Edge TTS, and WebSocket communication for responsive user interaction
- Conducted an empirical user study evaluating avatar-based CBT interventions across engagement, perceived support, communication effectiveness, and user preference metrics using Likert-scale questionnaires and quantitative statistical analysis
- Paper accepted for presentation at IEEE GEM 2026 in Berlin, in the session on Game AI, Generation and Development Tools
- Technologies: Unity, Python backend, Llama-3.2, text-to-speech synthesis, 3D avatar animation, WebSocket communication, user study evaluation, statistical analysis

DV1566 Cloud Computing Project – Serverless Image Processing on AWS Lambda

A BTH course project (3-member team) building a serverless image processing pipeline on AWS Lambda and analyzing its scaling behavior using AWS CloudWatch.

- Designed a serverless architecture where images uploaded to an Amazon S3 input bucket automatically trigger an AWS Lambda function that converts the image to grayscale and writes the result to a separate S3 output bucket
- Implemented the Lambda function in Node.js 24.x using AWS SDK v3 for S3 operations and the Sharp image-processing library, packaged as a Lambda Layer to reduce deployment size and improve reusability
- Built idempotent processing logic that checks the output filename prefix to skip already-processed images, preventing infinite trigger loops between the input and output buckets
- Analyzed performance using AWS CloudWatch metrics, observing a peak of 148 invocations in a short window out

of 323 total, an average execution duration of 263 ms (29 ms minimum, 1,013 ms maximum), and 2 errors across all executions

- Demonstrated automatic horizontal scale-out under increased upload load and scale-in as load decreased, based on rising and falling concurrent Lambda executions shown in CloudWatch
- Tools: AWS Lambda, Amazon S3, AWS CloudWatch, Node.js, Sharp, AWS SDK v3

GaiaOS – AI Orchestration Platform for Water, Energy, and Climate Infrastructure

A conceptual AI orchestration platform for unified management of water, energy, and climate infrastructure through a shared digital twin.

- Designed a cross-domain ecosystem optimization approach rather than siloed per-domain optimization across water, energy, and climate systems
- Created a cinematic demo video for the concept, initially built as a 12-scene animated experience in HTML, CSS, and JavaScript running over three minutes
- Rebuilt the demo using HTML5 Canvas with photorealistic visuals and real-time fluid and physics-based animations, including refined pause and resume behavior for a smoother presentation
- Technologies: HTML5 Canvas, JavaScript, CSS animations, physics-based simulation

Production Text Classification System – Cuisine Classification

End-to-end machine learning project classifying restaurant cuisines from a large-scale dataset.

- Designed, trained, and deployed a cuisine classification model achieving 96% accuracy on a 70,000-plus restaurant dataset, taking full ownership of experimentation, validation, and deployment
- Built scalable data preprocessing pipelines using Python and Pandas, with proper testing and documentation throughout the project lifecycle
- Tools: Python, scikit-learn, Pandas, Jupyter Notebooks – GitHub: github.com/Yasoda-Kishore/Machine-Learning-Internship

Production ML Regression System – Restaurant Rating Prediction

An end-to-end regression model predicting restaurant ratings from structured restaurant data.

- Built a regression model achieving an R squared of 0.9859 (98.59% variance explained), RMSE of 0.1792, and MAE of 0.1210 on a large-scale production dataset
- Implemented feature engineering and model optimization, achieving a MAPE of 3.74%, demonstrating strong generalization on unseen data
- Deployed the model with monitoring and continuous improvement workflows to maintain reliable production performance
- Tools: Python, scikit-learn, TensorFlow, Pandas, regression analysis

Restaurant Recommendation System – Content-Based Filtering

A content-based recommendation system built on a restaurant dataset with feature engineering for user preference modeling.

- Developed a content-based recommendation system on a 5,000-plus restaurant dataset using Python and scikit-learn, implementing feature engineering for user preference modeling and restaurant similarity analysis
- Achieved 42% weighted accuracy on a multi-class recommendation task across diverse restaurant categories, with weighted precision and recall between 0.38 and 0.42
- Implemented the filtering pipeline with proper train and test validation and documented results for stakeholder review
- Tools: Python, scikit-learn, Pandas, content-based filtering

Location-Based Analytics Dashboard

An interactive dashboard analyzing restaurant distribution, ratings, and location-based trends across cities.

- Built an interactive Streamlit dashboard with geospatial visualizations using Folium and interactive charts using Matplotlib and Seaborn
- Applied data quality validation and preprocessing pipelines to ensure accuracy and consistency across dashboard components, supporting data-driven business decisions
- Tools: Python, Streamlit, Folium, Pandas, Matplotlib, Seaborn

ACHIEVEMENTS & AWARDS

- **IEEE GEM 2026 Accepted Paper (2026):** “Does Embodiment Matter? Evaluating AI Avatars for CBT-Inspired Interventions in Games,” accepted for presentation at IEEE GEM 2026, Berlin
- **Oracle Data Science Professional Certification:** Covers foundational and applied data science workflows, including data preparation, model building, and model evaluation using Oracle Cloud Infrastructure Data Science tools
- **Google Advanced Data Analytics Certification:** Covers the full data analytics lifecycle, including data cleaning, exploratory analysis, statistical analysis, regression modeling, and machine learning fundamentals using Python

LANGUAGE SKILLS

- **English:** Professional Working Proficiency **Swedish:** Basic Level Proficiency **Telugu:** Native Language